

Fluid Flow Through Fluidized Bed

Pre-Lab Plan Unit Operations Lab #1

January 20th, 2026

Group 2, Tuesday, PM section

Srija Bomma

Lindsey Meisch

Kendall Johnson

Christian Salinas Rojas

1. Experimental objective

The objective of this experiment is to study the fluidization behaviour of solid particles by measuring the pressure drop across a bed as a function of the gas velocity. The experiment aims at finding the least fluidization velocity and comparing the experimental finding with the theoretical values such as the Ergun equation and the empirical fluidization correlations. .

2. Experimental

The purpose of the experiment is to examine how solid particles in a gas-solid system fluidize through the measurement of the pressure drop across a particle bed with variation of the superficial gas velocity. The experiment focuses on the air flow over the bed of silica sand and ion exchange resin particles in order to know the minimum fluidization velocity and to observe the effect of bed expansion and hysteresis. The experimental set up will comprise a plexiglass fluidized bed column, air distributor plate, rotameters, and a manometer to measure the pressure. The experiment, by providing a pressure drop and flow rate measurement at controlled conditions, compares measured variables to theoretical relationships, including the Ergun equation and experimental fluidization correlations so that experimental results may be compared with theoretical predictions of fluidized bed behavior.

a. Apparatus

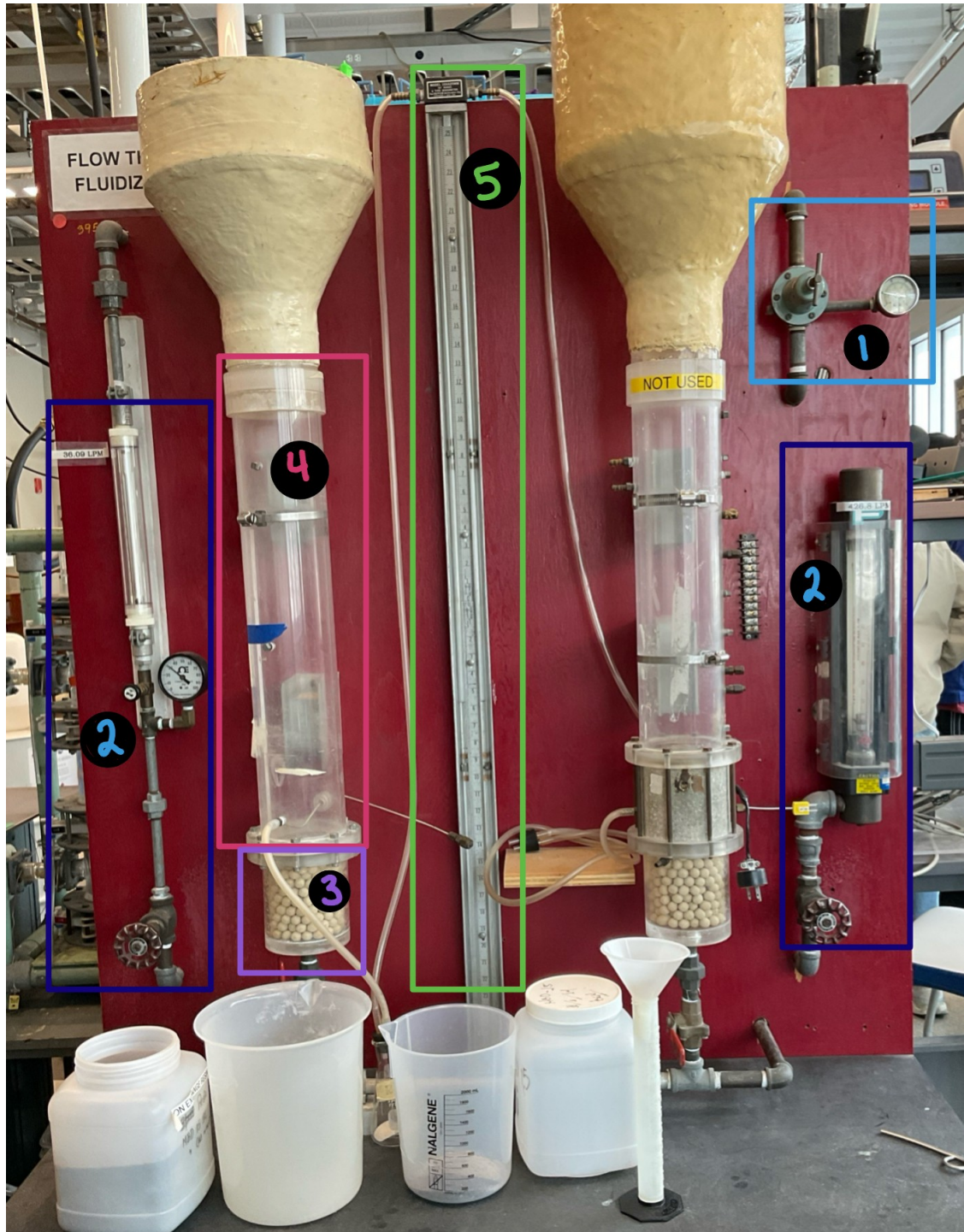


Figure 1: Apparatus

Table 1: Apparatus parts

Number/Color	Apparatus Part
1 - Light Blue	Compressed air inlet gauge
2 - Dark Blue	Rotameter(s) to control fluid flow through the column
3 - Purple	Air distributor plate
4 - Pink	Plexiglass fluidized bed column
5 - Light Green	Manometer

b. Materials and supplies

Table 2: Materials and supplies, and what their purpose is.

Materials/Supplies	Purpose/Specifications
Ion Exchange Resin	Medium to be put into the column
Sand	Medium to be put into the column
Balance	To weight sieves before/after passing medium through them to get a size distribution
Graduated Cylinder	Used to find the void fraction for the respective mediums
Funnel	Used to get the respective mediums into the
Compressed air	Sent through the column and medium
Vacuum cleaner(s)	There are two - one per medium - used to clean up the mediums
Sieves	Used to filter through the medium to determine the size distributions

c. Experimental plan

- [Prelab] Select seven sieves between ~500-100 micron that would span the expected particle size range.
- Weigh the seven sieves to obtain and record the tare masses.
- Stack the sieves from the highest micron to the lowest.

- Weigh out approximately 100g of resin and add to the top sieve and shake the stack of sieves to have the particles fall through.
- Weigh the sieves again to obtain the mass of the particles.
- Compute the mass fractions for each size and calculate the average particle diameter using the reciprocal-weighted average described in the Lab Manual.
- Repeat the same sieve procedure with the sand particles.
- Plot the mass fraction vs. particle size to determine and document the particle size distribution.
- Charge the 4" plexiglass fluidized-bed column with solids to a packed height of 6".
- Turn on the compressed air and start the air flow at a very low flow.
- Increase the flow in small, steady increments and record the rotameter flow rate, manometer pressure drop, and qualitative bed behavior at each step.
- Continue increasing the air flow until the bed is clearly beyond minimum fluidization, which should be identified when the pressure drop stabilizes and no longer increases with the increasing flow.
- Decrease the flow stepwise back to zero and record the same measurements at matching flow settings to capture hysteresis between the up-curve and down-curve.
- Repeat the same increasing-then-decreasing sequence with a packed height of 10" for the same material.
- Perform the above procedure for sand and resin.

The independent variables in this experiment include the superficial gas velocity ($\underline{V}_0$), varied from near 0 up to a value clearly above minimum fluidization. Bed material (sand vs. ion-exchange resin), initial packed bed height (6" and 10"), and flow direction (increasing vs. decreasing flow) are all included in the independent variables. The dependent variables are the pressure drop across the bed (ΔP)(from the manometer), normalized pressure gradient ($\Delta P/L$) (computed using measured ΔP and bed height), and the minimum fluidized velocity ($\underline{V}_{0M}$)(from the “knee/plateau” in ΔP vs. $\underline{V}_0$ and/or the intersection of packed-bed trend with plateau behavior).

The measurement range for $\underline{V}_0$ will be sampled from ~ 0 to “well above $\underline{V}_{0M}$ ” as indicated by the pressure-drop plateau. Step sizes are expected to be smaller near the expected transition region to better resolve $\underline{V}_{0M}$.

Two replicate readings of ΔP at each flow setpoint (taken back-to-back once the system stabilizes) will be taken. Report mean will aim for a $\pm 95\%$ confidence interval for ΔP at each setpoint. Uncertainty will be propagated into $\underline{V}_0$ and ΔP to estimate uncertainty in $\Delta P/L$ and $\underline{V}_{0M}$. The manometer will be calibrated with a zero check before each material run to ensure no-flow ($\Delta P \approx 0$). A rotameter consistency check will be performed by returning to one mid-range flow setpoint at the end of a run and the reading will be confirmed to match the earlier value within instrument resolution. For the packed-bed regime fits, RMSE and R^2 will be used between measured and predicted $\Delta P/L$ vs $\underline{V}_0$ below $\underline{V}_{0M}$.

Rotameter readings will be converted to volumetric flowrate and then to superficial velocity using the column cross-sectional area. $\Delta P/L$ vs $\underline{V}_0$ plotted for each case, showing both increasing and decreasing curves to visualize hysteresis. Identify the experimental $\underline{V}_0$ from the

transition where ΔP stops increasing significantly with velocity. In the packed-bed regime, compare the measured $\Delta P/L$ to the Ergun equation using the measured particle size. Finally, compute the predicted minimum fluidization velocity using the correlation referenced in the Lab Manual and compare to the experimental value.

3. Project safety evaluation

Overall the lab does not present any significant danger. There is no chemical reaction so there is no heat generation or gas evolution. The two main sources of hazards are the compressed air and the solid material particles used. Compressed air presents the possibility of unwanted escape. This is unlikely but could prove to be dangerous if there is failure in containment. The solid particles of both sand and ionic resin can prove themselves to be irritants if handled improperly. There is also a risk of slight inhalation as the fluidization occurs, however most of these particles are too large to be carried out of the column. Finally, due to their size they can also pose a slip hazard. It is important to “clean-as-you-go” so no significant amount of waste accumulates on the floor.

	<i>Yes/No</i>	
Potential electrical hazards?	N	If yes, identify conditions:
Potential mechanical hazards?	N	If yes, identify conditions:
Potential pressure hazards?	Y	If yes, identify conditions: High pressure from compressed air
Potential temperature hazards?	N	If yes, identify conditions:
Hazardous/toxic chemicals involved?	N	If yes, hazards involved:
Gloves required?	N	If yes, specify type:
MSDS reviewed by all team members?	N	List MSDS sheets reviewed:
<i>Process Parameters</i>	<i>Max Expected</i>	<i>List location and possible hazards and precautions</i>
Temperature (°C)	RT	
Pressure (psia)	134.7	Compressed air in, regulated down
<i>Safety Controls</i>	<i>yes/no</i>	<i>If yes, Provide description</i>
	Y	Shut off and release valve for air

I have read relevant background material for the Unit Operations Laboratory entitled: " Fluid Flow Through Fluidized Bed " and understand the hazards associated with conducting this experiment. I have planned out my experimental work in accordance to standards and acceptable safety practices and will conduct all of my experimental work in a careful and safe manner. I will also be aware of my surroundings, my group members, and other lab students, and will look out for their safety as well.

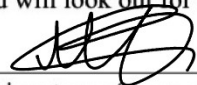
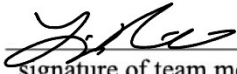
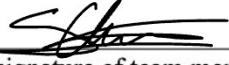
	1/20/26
signature of team member	date
	1/20/26
signature of team member	date
	1/20/2026
signature of team member	date
	1/20/26

Figure 2: Signed safety chart